

Machine Learning Model Comparison & Evaluation Report

Task 2: Build and Compare Classification Models

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Dataset: Breast Cancer Diagnostic Dataset (scikit-learn)

1. Executive Summary

This project explores the implementation, training, and evaluation of three distinct machine learning classification algorithms: **Logistic Regression**, **Decision Tree**, and **Random Forest**. Using the standard Breast Cancer Wisconsin dataset, the objective is to evaluate model efficacy in predicting whether a tumor is malignant or benign, relying on key performance metrics such as accuracy, precision, recall, and F1-score.

2. Dataset Overview

- **Source:** Built-in scikit-learn dataset (Breast Cancer Diagnostic Dataset).
- **Features:** 30 numeric attributes computed from digitized images of fine needle aspirates (FNA) of breast masses (radius, texture, perimeter, area, smoothness, etc.).
- **Target Variable:** Binary classification label where 0 represents Malignant and 1 represents Benign.
- **Data Preprocessing:** Missing values verified, features normalized using `StandardScaler` to achieve zero mean and unit variance, and data split into 80% training and 20% testing subsets.

3. Models Implemented

- **Logistic Regression:** A linear baseline model used for binary classification, optimal for estimating the probability of discrete class outcomes.
- **Decision Tree Classifier:** A non-linear tree-structured model that recursively partitions data based on feature thresholds to segment classes.
- **Random Forest Classifier:** An ensemble learning method combining multiple decision trees via bootstrap aggregation to control overfitting and maximize generalization accuracy.

4. Evaluation Metrics

Models were rigorously evaluated on the held-out test set using the following performance indicators:

- **Accuracy:** Proportion of correct predictions among total instances.
- **Precision:** Proportion of true positive identifications out of all predicted positive outcomes.
- **Recall (Sensitivity):** Proportion of actual positives correctly identified by the model.

- **F1-Score:** Harmonic mean balancing precision and recall.
- **Confusion Matrix:** Detailed breakdown of true positives, false positives, true negatives, and false negatives.

5. Results & Major Findings

All three models achieved exceptional overall accuracy exceeding 95% due to the clean separation within the feature space. **Random Forest** demonstrated superior resilience and stability by mitigating variance through ensemble averaging. **Logistic Regression** functioned as an efficient, rapid linear benchmark, while the **Decision Tree** captured complex feature splits effectively.

6. Conclusion

The comparative analysis confirms that ensemble architectures like Random Forest are exceptionally well-suited for medical diagnostic classification tasks by reducing critical misclassification risks. The complete pipeline, source code, visual reports, and documentation have been successfully packaged and submitted to GitHub.